

Micro-3D-Printed Antenna-in-Package Substrates with Quasi-Coaxial Through Vias

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Novelty of work

The key innovation of this work is the development of a quasi-coaxial through-via-embedded packaging substrate for antenna-in-package (AiP) applications at 28 GHz, eliminating the need for traditional molding and grinding processes by integrating 3D printing. This additive manufacturing approach significantly reduces both the cost and complexity of fabricating AiP devices.

Abstract Text

Antenna-in-package (AiP) technology integrates antennas directly into the semiconductor package, offering significant improvements in signal integrity and system performance [1, 2]. This integration not only reduces the overall device footprint but also enhances the antenna's performance by leveraging the controlled environment within the package. AiP is becoming increasingly essential in the development of compact, high-performance wireless communication devices [3, 4]. In conventional AiP designs, before a patch antenna is stacked on top of a molded flip-chip die, a through-mold via (TMV) must be formed to connect the signal line to the chip (Fig. 1a). However, this TMV process is costly, requiring multiple steps including TMV etching, molding, and substrate grinding. In this research, we propose a 3D-printed AiP substrate composed of quasi-coaxial vias, a cavity for a chip, and a patch antenna (Fig. 1b). By utilizing 3D printing, the costs associated with TMV etching and interposer assembly can be eliminated.

In this work, projection micro stereolithography (PμSL) technology (BMF) was employed, and a 4-by-4-mm substrate was printed with a 500-μm thickness and a 100-μm via diameter. The signal via was designed to be surrounded by ground vias, i.e., quasi-coaxial vias (Fig. 1c). The substrate exhibited a dielectric constant of 2.4 and a loss tangent of 0.043 at 10.3 GHz. The surface of the printed polymer was hydrophilized through low-temperature atomic layer deposition, enabling electroless plating to form a metal seed layer. To implement the quasi-coaxial vias, CST simulations were used to optimize the via count and the spacing between the signal and ground vias. A patch antenna was designed to match the resonance frequency to 28 GHz, verifying the AiP assembly's functionality. Electrical characterization of the fabricated AiP was performed using a network analyzer, and the propagation pattern was analyzed in an anechoic chamber. This work demonstrates that the 3D-printed substrate can substitute the molding layer by embedding

quasi-coaxial through vias and a chip cavity, achieving a cost-effective and simplified fabrication process.

References

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Figures

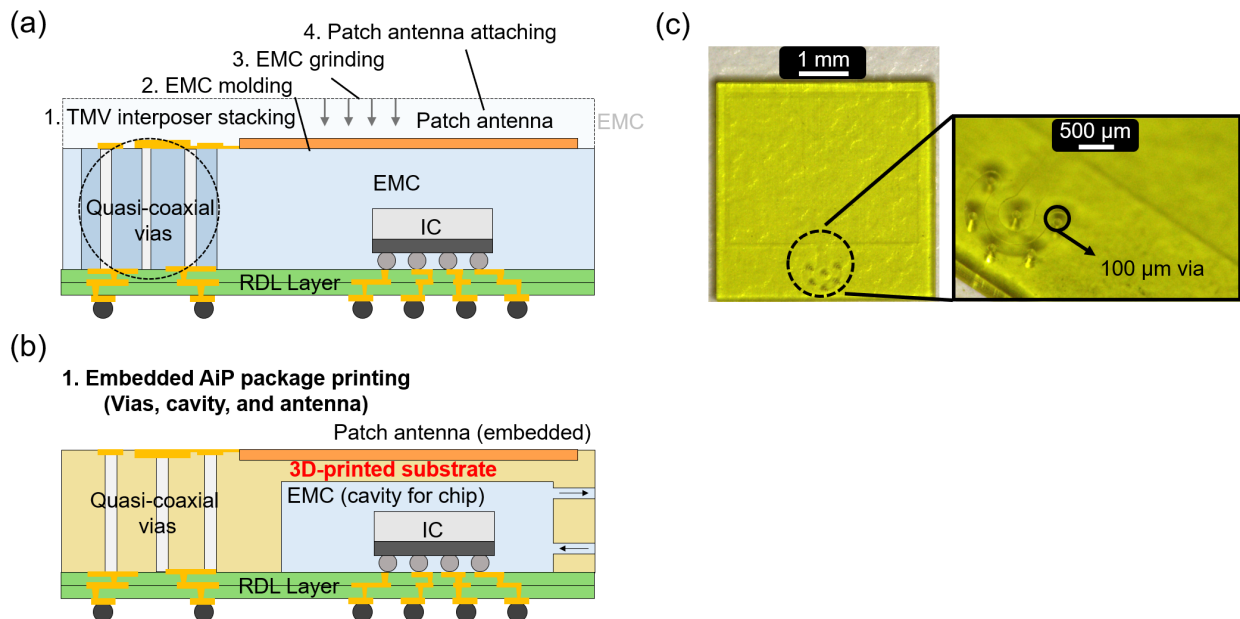


Figure 1. (a) Comparison between a conventional and 3D-printed substrate. (b), (c) The illustration and images of through via-embedded AiP with patch antenna, respectively.